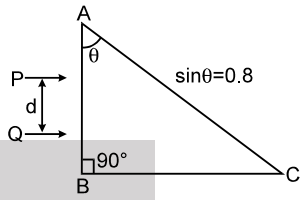
High Level Problems (HLP)

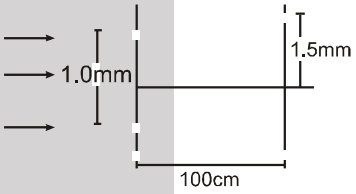
SUBJECTIVE QUESTIONS

1. Two parallel beams of light P & Q (separation d) containing radiations of wavelengths 4000 Å & 5000 Å (which are mutually coherent in each wavelength separately) are incident normally on a prism as shown in figure. The refractive index of the prism as a function of wavelength is given by the relation, $\mu(\lambda) = 1.20 + \frac{b}{\lambda^2}$, where λ is in Å & b is a positive constant. The value of b is such that the condition for total reflection at the face AC is just satisfied for one wavelength & is not satisfied for the other. A convergent lens is used to bring these transmitted beams into focus. If the intensities of the upper & the lower beams immediately after transmission from the face AC, are 4I & I respectively, find the resultant intensity at the focus.



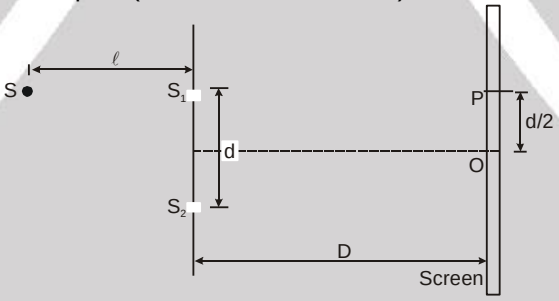
[JEE 1991, 8]

2. White coherent light (400 nm-700 nm) is sent through the slits of a Young's double slit experiment (as shown in the figure). The separation between the slits is 1 mm and the screen is 100 cm away from the slits. There is a hole in the screen at a point 1.5 mm away (along the width of the fringes) from the central line. (a) For which wavelength(s) there will be minima at that point ? (b) which wavelength(s) will have a maximum intensity?



3. A beam of light consisting of two wavelengths, 6500 Å and 5200 Å is used in double slit experiment ($1\ Å = 10^{-10}$ m). The distance between the slits is 2.0 mm and the distance between the plane of the slits and the screen is 120 cm. (a) Find the distance of the third bright fringe on the screen from the central maximum for the wavelength 6500 Å. (b) What is the least distance from the central maximum where the bright fringes due to both the wavelengths coincide?

4. A source S is kept directly behind the slit S_1 in a double–slit apparatus. Find the phase difference at a point O which is equidistant from S_1 & S_2. What will be the phase difference at P if a liquid of refraction index μ is filled; (wavelength of light in air is λ due to the source). Assume same intensity due to S_1 and S_2 on screen and position at liquid. ($\lambda << d$, $d << D$, $\ell >> d$)

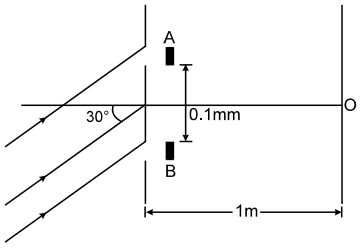


(a) between the screen and the slits.
(b) between the slits & the source S. In this case find the minimum distance between the points on the screen where the intensity is half the maximum intensity on the screen.

5. A monochromatic light of λ = 5000 Å is incident on two slits separated by a distance of 5×10^{-4} m. The interference pattern is seen on a screen placed at a distance of 1 m from the slits. A thin glass plate of thickness 1.5×10^{-6} m & refractive index μ = 1.5 is placed between one of the slits & the screen. Find the intensity at the centre of the screen, if the intensity there is I_0 in the absence of the plate. Also find the lateral shift of the central maximum. **[REE 1993, 4]**

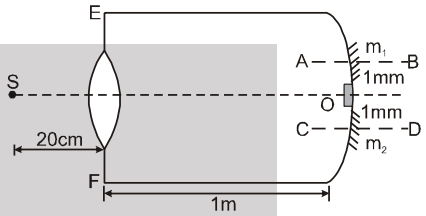
6. In a YDSE experiment, the distance between the slits & the screen is 100 cm. For a certain distance between the slits, an interference pattern is observed on the screen with the fringe width 0.25 mm. When the distance between the slits is increased by Δd = 1.2 mm, the fringe width decreased to n = 2/3 of the original value. In the final position, a thin glass plate of refractive index 1.5 is kept in front of one of the slits & the shift of central maximum is observed to be 20 fringe width. Find the thickness of the plate & wavelength of the incident light.

7. In a YDSE a parallel beam of light of wavelength 6000 Å is incident on slits at angle of incidence 30º. A & B are two thin transparent films each of R.I. 1.5. Thickness of A is 20.4 μm. Light coming through A & B have intensities I and 4I respectively on the screen. Intensity at point O which is symmetric relative to the slits is 3 I. The central maxima is above O.



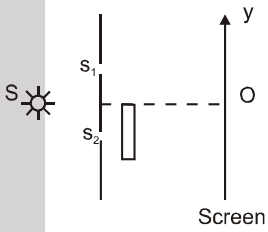
(a) What is the maximum thickness of B to do so.
Assuming thickness of B to be that found in part (a) answer the following parts.
(b) Find fringe width, maximum intensity & minimum intensity on screen.
(c) Distance of nearest minima from O.
(d) Intensity at 5 cm on either side of O.

8. An equi convex lens of focal length 10 cm (in air) and R.I. 3/2 is put at a small opening on a tube of length 1 m fully filled with liquid of R.I. 4/3. A concave mirror of radius of curvature 20 cm is cut into two halves m_1 and m_2 and placed at the end of the tube. m_1 & m_2 are placed such that their principal axis AB and CD respectively are separated by 1 mm each from the principal axis of the lens.



A slit S placed in air illuminates the lens with light of frequency 7.5×10^{14} Hz. The light reflected from m_1 and m_2 forms interference pattern on the left end EF of the tube. O is an opaque substance to cover the hole left by m_1 & m_2. Find :
(a) the position of the image formed by lens water combination.
(b) the distance between the images formed by m_1 & m_2.
(c) width of the fringes on EF.

9. The Young's double slit experiment is done in a medium of refractive index 4/3. A light of 600 nm wavelength is falling on the slits having 0.45 mm separation. The lower slit S_2 is covered by a thin glass sheet of thickness 10.4 μm and refractive index 1.5. The interference pattern is observed on a screen placed 1.5 m form the slits as shown. **[JEE 1999 (Main), 5+3+2/200]**



(a) Find the location of central maximum (bright fringe with zero path difference) on the y-axis.
(b) Find the light intensity at point O relative to the maximum fringe intensity.
(c) Now if 600 nm light is replaced by white light of range 400 nm to 700 nm, find the wavelengths of the light that form maxima exactly at point O.
[All wavelengths in the problem are for the given medium of refractive index 4/3. Ignore dispersion]

10. A glass plate of refractive index 1.5 is coated with a thin layer of thickness t and refractive index 1.8. Light of wavelength λ travelling in air is incident normally on the layer. It is partly reflected at the upper and the lower surface of the layer and the two reflected rays interfere. Write the condition for their constructive interference. If λ = 648 nm, obtain the least value of t for which the rays interfere constructively. **[JEE 2000 (Main), 4/100]**

HLP Answers

1. 9I

2. (a) 428 nm, 600 nm, (b) 500 nm

3. (a) 1.17 mm. (b) 1.56 mm

4. (a) $\Delta\phi = \left(\frac{1}{\ell} + \frac{\mu}{D}\right)\frac{\pi d^2}{\lambda}$ (b) $\Delta\phi = \left(\frac{\mu}{\ell} + \frac{1}{D}\right)\frac{\pi d^2}{\lambda}$; $D_{min} = \frac{\beta}{2} = \frac{\lambda D}{2d}$

5. 0, 1.5 mm

6. λ = 600 nm , t = 24 μm

7. (a) t_B = 120 μm (b) β = 6 mm ; $I_{max} = 9I$, $I_{min} = I$ (c) β/6 = 1 mm
(d) I (at 5 cm above O) = 9I, I (at 5 cm below O) = 3 I

8. (a) 80 cm behind the lens (b) 4 mm (c) β = 60 μm

9. (a) y = – 4.33 mm (b) $I_0 = \frac{3}{4} I_{max}$ (c) λ = 650 nm, 433.3 nm

10. $2\mu t = \left(n + \frac{1}{2}\right)\lambda$ with μ = 1.8 and n = 0,1,2,3..... ; 90 nm